

TELECOM CUSTOMER INSIGHTS

Customer Churn & Retention Analysis

Identifying churn drivers and data-driven retention opportunities across 7,043 customers

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BRAC University | Data source: IBM Telco Customer Churn dataset

Objective & methodology

Customer churn is one of the most expensive problems telecom operators face — acquiring a new customer typically costs far more than retaining an existing one.

This analysis identifies who is leaving, why, and what actions could reduce it.

01

Data cleaning

Standardized fields, resolved missing values, verified data types

02

Exploratory analysis

Churn examined across contract, tenure, pricing, and services

03

Customer segmentation

Customers grouped into four behavior-based risk segments

04

Business recommendations

Findings translated into targeted retention actions

DATASET AT A GLANCE

7,043

Customer records analyzed

21

Attributes per customer (contract, billing, services, tenure)

26.5%

Overall churn rate

4

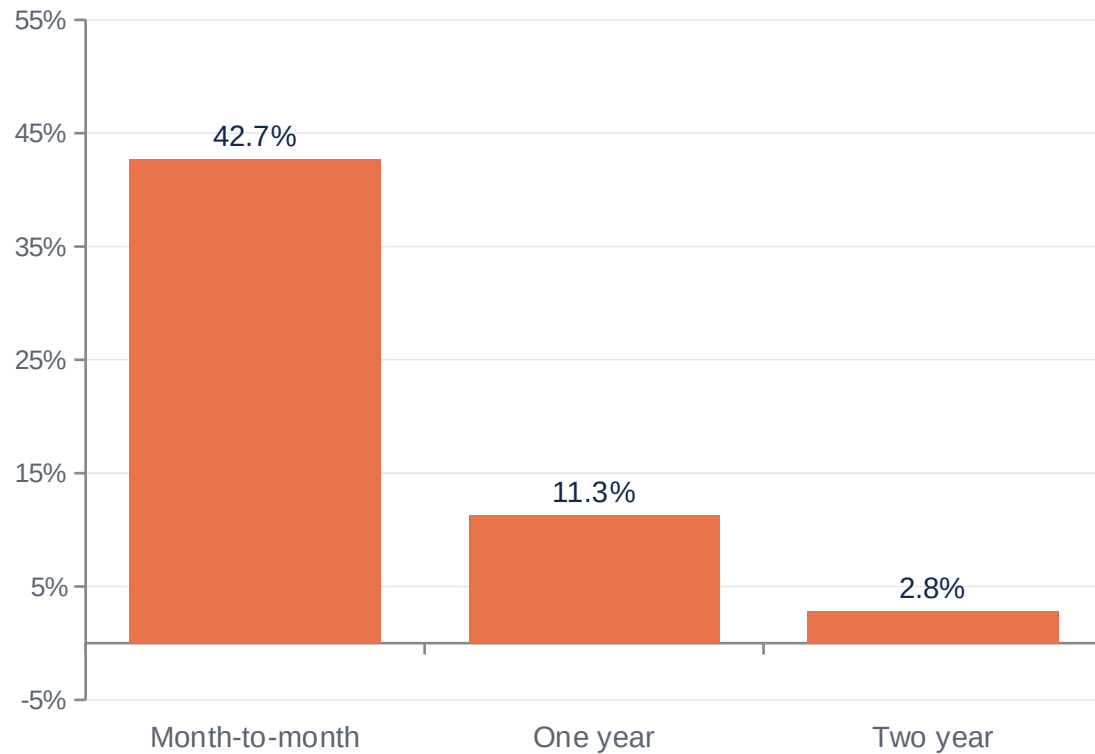
Behavior-based customer segments identified

KEY FINDING 01

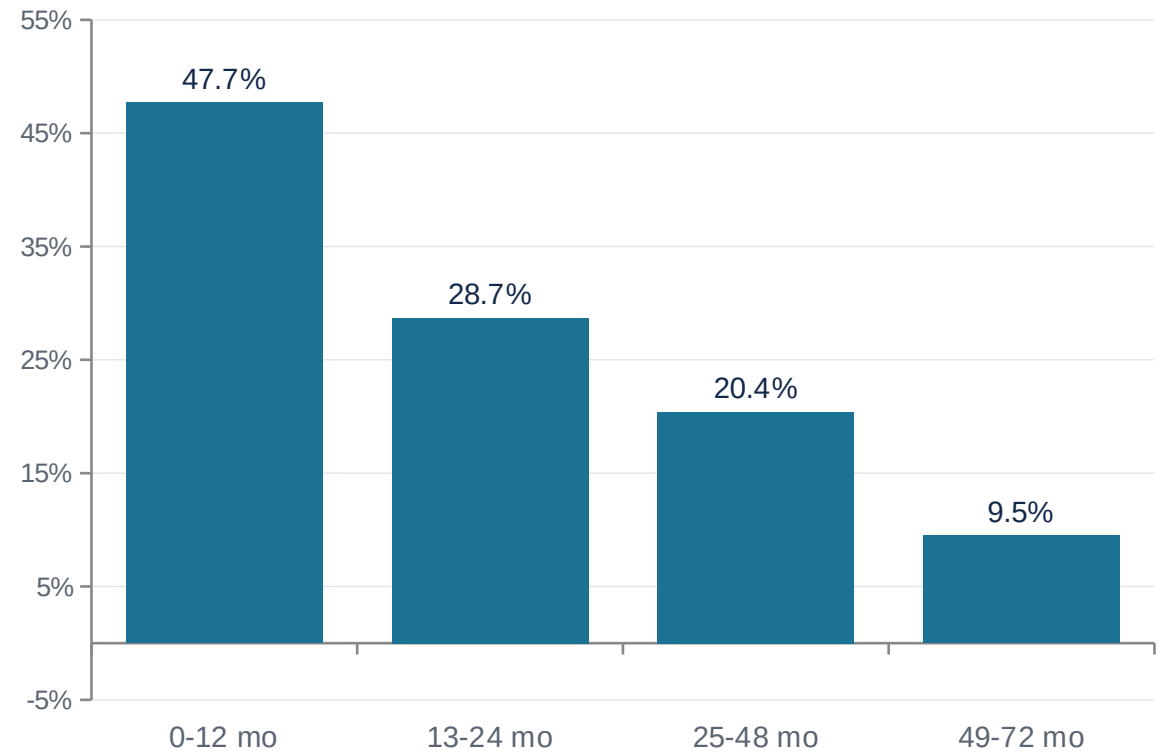
Commitment matters more than price

Customers without a contract lock-in, and those still new, churn at dramatically higher rates.

Churn rate by contract type



Churn rate by customer tenure

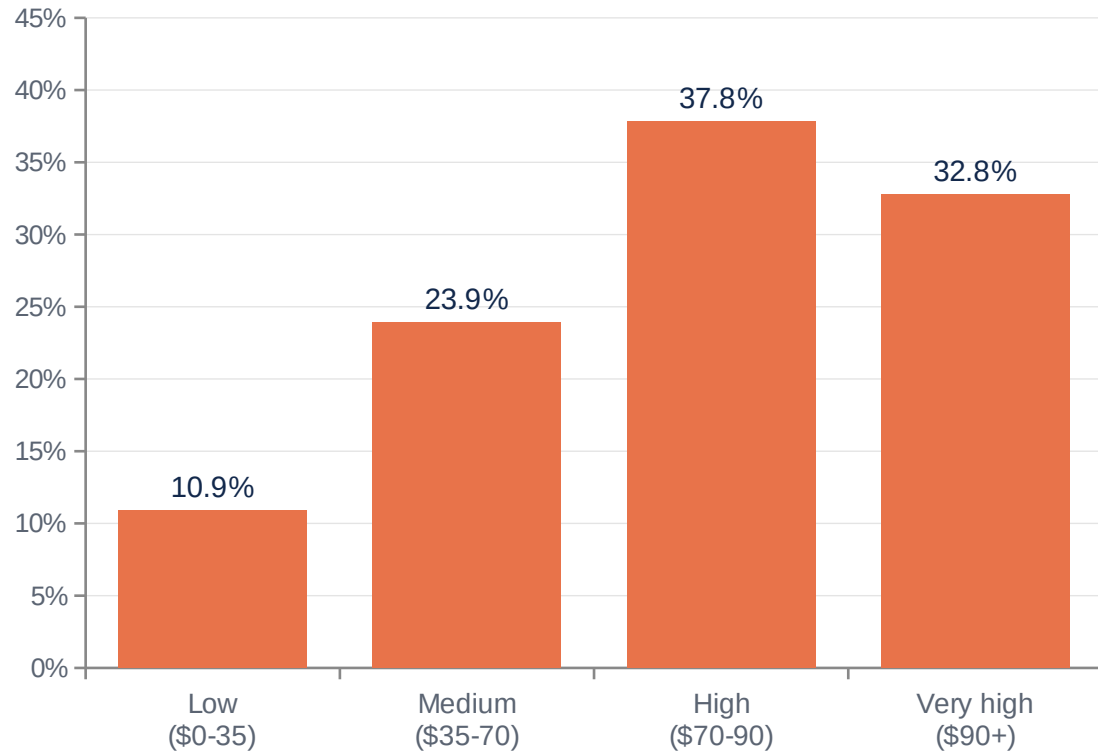


KEY FINDING 02

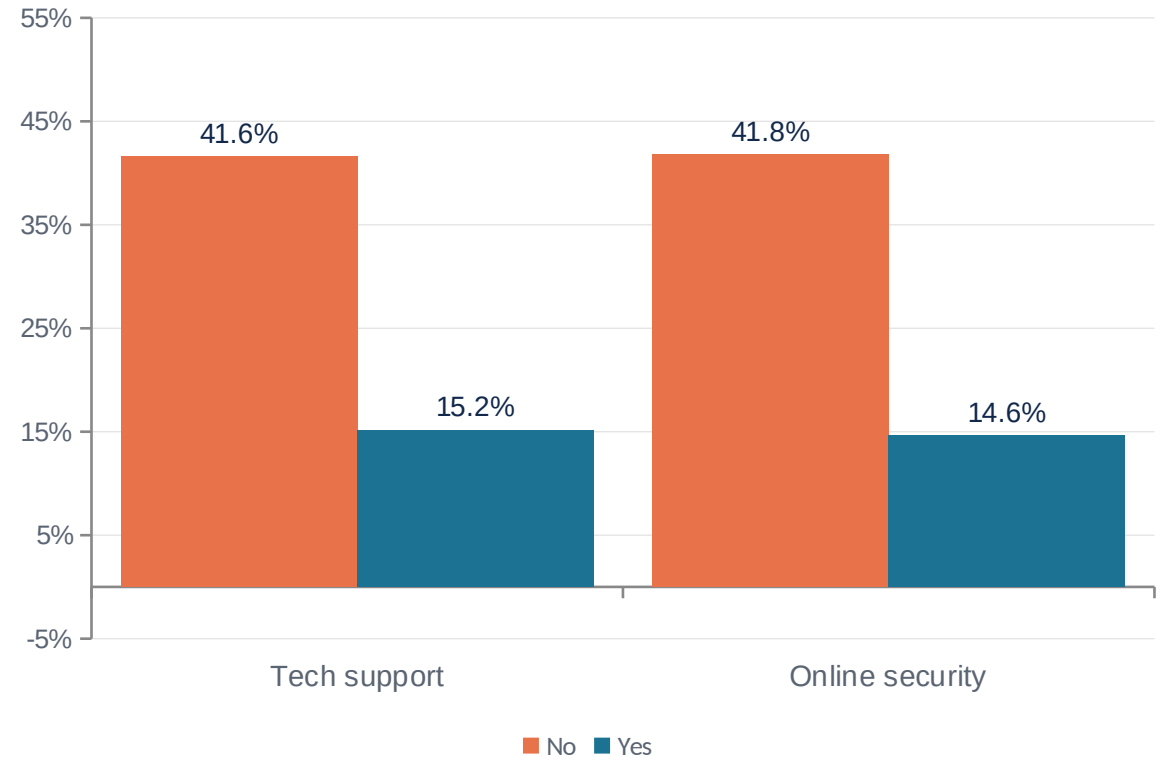
Support services strongly reduce churn

Higher bills raise churn risk, but bundled support and security services cut it by more than half.

Churn rate by monthly charges



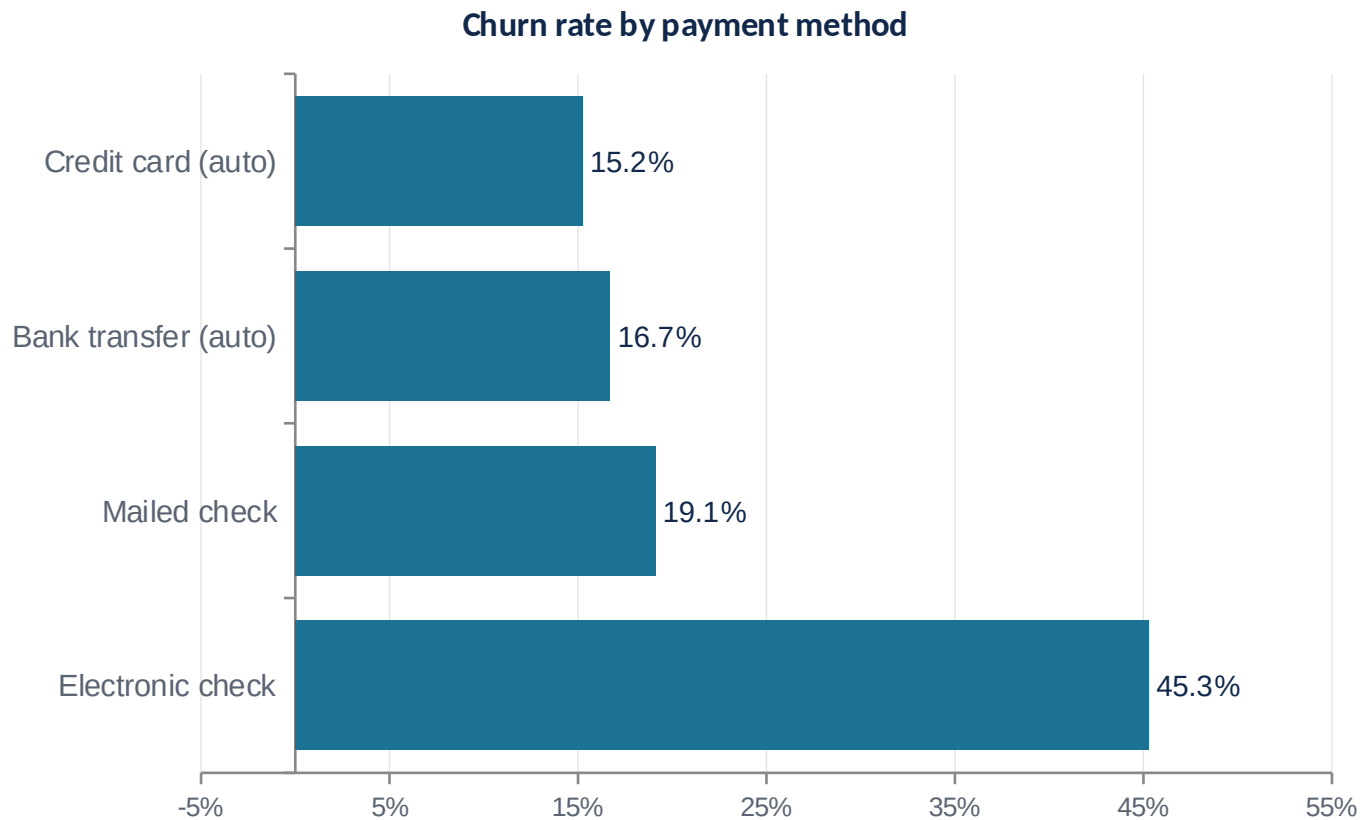
Churn rate: with vs without services



KEY FINDING 03

Payment behavior signals engagement

Customers on manual payment methods churn nearly 3x more than those on autopay.



WHY IT MATTERS

Electronic check

45.3% churn

Manual, one-off payments correlate with lower commitment and weaker engagement — a simple flag for retention outreach.

Autopay methods (bank transfer, credit card) churn at 15-17% — under half the rate.

Four customer segments, four different risk levels

Customers grouped by contract type, tenure, pricing, and support services.

High-risk new

58.96%

churn rate

1,545 customers

4.6 mo avg tenure

\$67.41 avg charge

Price-sensitive at-risk

43.44%

churn rate

1,151 customers

34.0 mo avg tenure

\$90.22 avg charge

Stable / moderate-risk

11.55%

churn rate

3,671 customers

37.4 mo avg tenure

\$51.64 avg charge

Loyal long-term

5.03%

churn rate

676 customers

65.8 mo avg tenure

\$86.61 avg charge

Turning insight into retention actions

1

Convert month-to-month customers to annual contracts

Incentivize the switch from 42.7% to 2.8% churn risk.

2

Strengthen onboarding in the first 12 months

New customers churn at 47.7% — the highest of any group.

3

Bundle support services for at-risk segments

Tech support & security cut churn from ~42% to ~15%.

4

Review pricing at the \$70-90 charge tier

This bracket shows the single highest churn rate, 37.8%.

5

Promote autopay adoption

Electronic check users churn 45.3% vs 15-17% on autopay.

6

Prioritize the two highest-risk segments

High-Risk New & Price-Sensitive At-Risk = 38% of customers.

CONCLUSION

Churn is predictable — and preventable

The data shows churn isn't primarily about price.

Even high-paying customers stay loyal (5% churn) when they have long tenure, contract commitment, and support services — while similarly high-paying customers without those show 43% churn.

26.5%

Current overall churn rate

2

Segments driving 65% of churn

6

Targeted retention actions identified

Full analysis, live dashboard, and code:

github.com/tanjina1261/telecom-customer-churn-analysis